

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms an integral to find mass.	AO1.1a	M1	$M = \rho \int_0^2 \frac{\pi x^4}{4} dx$ $= \pi \rho \left[\frac{x^5}{20} \right]_0^2$ $= \frac{8}{5} \pi \rho$ $\bar{x} \times \frac{8}{5} \pi \rho = \int_0^2 \frac{\pi \rho x^5}{4} dx$ $= \pi \rho \left[\frac{x^6}{24} \right]_0^2$ $= \frac{8}{3} \pi \rho$ $\bar{x} = \frac{5}{3}$
	Obtains correct mass.	AO1.1b	A1	
	Forms an integral of the form $\int xy^2 dx$	AO1.1a	M1	
	Obtains correct value of $\int xy^2 dx$	AO1.1b	A1	
	Obtains correct centre of mass, follow through 'their' mass provided both M1 marks have been awarded.	AO1.1b	A1F	
(b)	Uses trigonometry and 'their' centre of mass to form equation to find angle.	AO3.1b	M1	$\tan \alpha = \frac{2}{2 - \frac{5}{3}}$ $\alpha = 80.5^\circ$
	Finds the correct angle to the required accuracy. FT 'their' values provided all M1 marks have been awarded	AO1.1b	A1F	
				Total 7 marks

Q2.

- (a) $\pi \int xy^2 dx = \pi \int_0^{2r} x(x + 3r)^2 dx$
(use of π must be consistent)
attempt to integrate $\int xy^2 dx$
- must involve three terms

M1

$$= \pi \int_0^{2r} (x^3 + 6rx^2 + 9r^2x) dx$$

$$= \pi \left[\frac{x^4}{4} + 2rx^3 + \frac{9r^2x^2}{2} \right]_0^{2r}$$

correct integration

A1

$$\pi[4r^4 + 16r^4 + 18r^4]$$

Limits used correctly

m1

$$38r^4\pi$$

correctly evaluated in terms of r^4

A1

$$x = \frac{38r^4\pi}{98\pi r^3/3} = \frac{57r}{49}$$

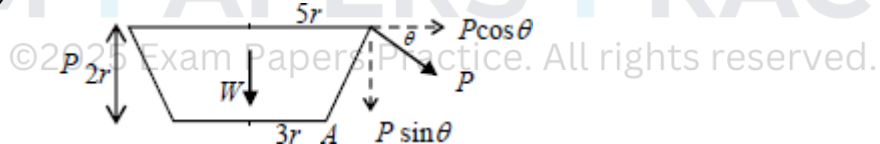
M1

$\frac{\pi \int xy^2 dx}{\text{use of volume}}$
 ft 'their' $\int xy^2 dx$

A1F

6

(b) (i)



Moments at A:

$$W(3r) = P \sin \theta (2r) + P \cos \theta (2r)$$

*M1 attempt at moments - evidence of
force $\times$ perpendicular distance*

A1 two terms correct

M1A1

A1 all terms correct

A1

$$3W = 2P(\cos \theta + \sin \theta)$$

$$\frac{3W}{2(\cos\theta + \sin\theta)} = P$$

AG Must see evidence of factorising

A1

4

- (ii) Min value of P is when $\cos\theta + \sin\theta$ is at a maximum.

Attempt to maximise denominator

M1

Max value of $\cos\theta + \sin\theta$ is $\sqrt{2}$
 $\sqrt{2}$ seen

A1

Max P value = $\frac{3W}{2\sqrt{2}}$

Or equiv eg $\frac{6\sqrt{2}W}{8}, \frac{3}{4}\sqrt{2}W$ etc

A1

3

- (iii) $\theta = 45^\circ$

B1

1

[14]

Q3.

(a) Volume of cone = $\frac{1}{3}\pi r^2 h = \frac{8h^3\pi}{3}$

Correct volume

B1

Alternative: $\left[\pi \int_0^{8h} \left(\frac{1}{8}x \right)^2 dx = \frac{8h^3}{3}\pi \right]$

(B1)

$$\pi \int xy^2 dx = \pi \int_0^{8h} x \left(\frac{x^2}{8} \right) dx = \pi \int_0^{8h} \frac{x^3}{64} dx$$

Use of $\int xy^2 dx$

M1

A1

$$= 16h^4$$

$$\bar{x} = \frac{\pi \int xy^2 \, dx}{\pi \int y^2 \, dx} = \frac{16h^4\pi}{8h^3\frac{\pi}{3}}$$

Attempt at formula for $\bar{x}$

M1

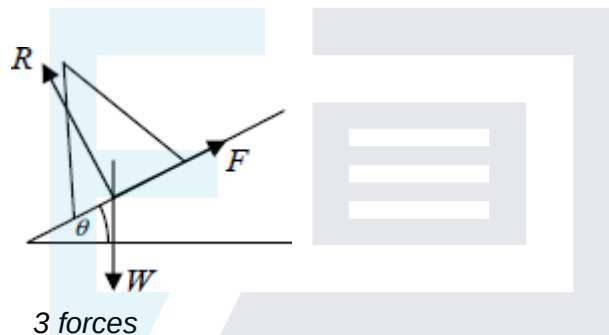
$$= 6h$$

AG

A1

5

(b) (i)

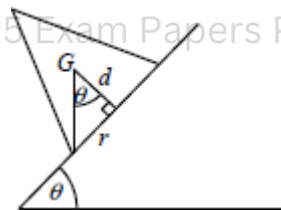


Through a common point at base of cone

B1
B1

2

(ii)



G above point of rotation

$$\tan \theta = \frac{r}{d}$$

tan θ seen anywhere

M1

$$= \frac{h}{8h - 6h} = \frac{1}{2}$$

Ratio correct

A1

$$\theta = \tan^{-1} \frac{1}{2} = 26.6^\circ$$

Use of $\tan^{-1}$ to get θ ; ft incorrect ratio

A1F

3

(c)
$$\left. \begin{array}{l} \text{Resolve perp to plane } N = W \cos \alpha \\ \text{Parallel to plane } F = W \sin \alpha \end{array} \right\}$$

Both correct

B1

Limiting friction
$$F = \mu N \Rightarrow \tan \alpha = \frac{6}{13}$$

Eliminate W , obtain $\tan \alpha$

M1

$$\alpha = 24.8^\circ < 26.6^\circ \quad \text{or} \quad \frac{6}{13} < \frac{1}{2}$$

Comparison

A1

$\Rightarrow$ slides first

Conclusion; ft (b)(ii)

A1F

4

[14]

Q4.

(a)
$$\int_0^{2r} xy^2 dx = \int_0^{2r} \frac{x^3}{4} dx$$

Attempt to use formula $\int xy^2 dx$

M1

$$= \left[\frac{x^4}{16} \right]_0^{2r}$$

$$= r^4$$

Integration correct

A1

$$\int_0^{2r} y^2 dx = \int_0^{2r} \frac{x^2}{4} dx$$



Or use of $\frac{1}{3} \pi r^2 h$ to get $\frac{2}{3} \pi r^3$

$$= \left[\frac{x^3}{12} \right]_0^{2r}$$

$$= \frac{2r^3}{3}$$

B1

$$\Rightarrow \bar{x} = r^4 \div \frac{2r^3}{3} = \frac{3r}{2}$$

$$\bar{x} = \frac{\pi \int_0^{2r} xy^2 dx}{\pi \int_0^{2r} y^2 dx}$$

AG use of

NB – consistent use of π throughout for M1A1 at end (or cancelled at start)

M1A1

5

(b) (i)

	mass	distance
Lower	$\pi r^2 (2r) \rho$	r
Upper	$\frac{\pi r^3}{3} (2r) k \rho$	$2r + \frac{r}{2}$

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Any correct pairing seen anywhere
(mass $\leftrightarrow$ distance)

B1

$$\left(\pi 2r^3 \rho + \frac{\pi 2r^3}{3} k \rho \right) \bar{x} = \pi 2r^3 \rho (r)$$

Equation formed

M1

$$+ \frac{\pi 2r^3}{3} k \rho \left(\frac{5r}{2} \right)$$

lose 1 each 'type' of error

A2,1,0

$$\Rightarrow \left(1 + \frac{k}{3}\right) \bar{x} = r + \frac{5rk}{6}$$

$$\Rightarrow (6 + 2k) \bar{x} = (6 + 5k) r$$

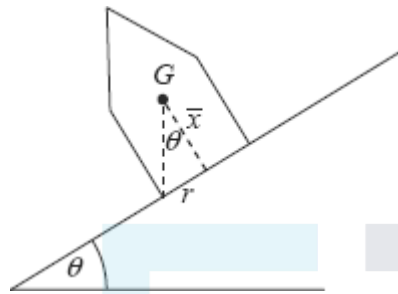
$$\bar{x} = \left(\frac{6 + 5k}{6 + 2k}\right) r$$

Rearrange to obtain printed answer

A1

5

(ii)



$$\tan \theta = \frac{r}{\bar{x}}$$

Use of tan theta

Correct structure

M1

A1

$$\Rightarrow \frac{2}{3} = \frac{r}{\left(\frac{6 + 5k}{6 + 2k}\right) r}$$

Substitution of $\bar{x}$, tan theta

B1

$$\frac{2}{3} = \frac{6 + 2k}{6 + 5k}$$

$$12 + 10k = 18 + 6k$$

Attempt to solve

M1

$$4k = 6$$

$$k = \frac{3}{2}$$

A1

5

[15]

Q5.

(a) $\text{volume} = \pi \int y^2 dx$

$$= \pi \int_0^2 (4 - x^2) dx$$

evidence of attempt at $\int y^2 dx$

M1

$$= \pi \left[4x - \frac{x^3}{3} \right]_0^2$$

integrating

A1

$$= \pi \left[8 - \frac{8}{3} - 0 \right]$$

$$= \frac{16\pi}{3}$$

AG

A1

3

(b) $\frac{16\pi}{3} \bar{x} = \pi \int_0^2 x(4 - x^2) dx$

$$= \pi \int_0^2 (4x - x^3) dx$$

attempt at $\int xy^2 dx$

M1

$$= \pi \left[2x^2 - \frac{x^4}{4} \right]_0^2$$

integrating correctly

A1

$$= \pi[8 - 4 - 0]$$

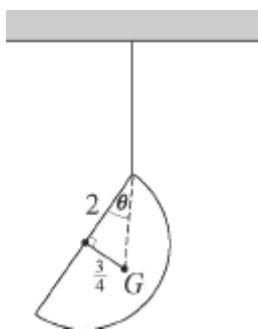
equation to find $\bar{x}$ (dependent on first M1)

m1

$$= 4\pi$$

$$\Rightarrow \bar{x} = \frac{3}{4}$$

(c)



$$\tan \theta = \frac{3}{4}$$

tan theta seen

$$= \frac{3}{8}$$

structure correct

$$\frac{3}{8}$$

M1

A1ft

$$\Rightarrow \theta = 20.6^\circ$$

accept AWWF 20° – 21°; ft their x

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A1ft

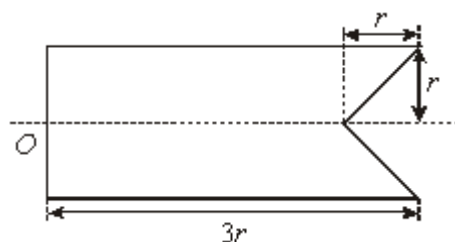
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[10]

Q6.

(a)



Cylinder

Weight

$$3\pi r^3 \rho$$

Dist from O

$$\frac{3r}{2}$$

Cone	$\frac{1}{3} \pi r^3 \rho$	$2r + \frac{3r}{4} = \frac{11r}{4}$
New body	$\frac{8}{3} \pi r^3 \rho$	$\bar{x}$

3 shapes

M1

-1EE

A2

$$\frac{8}{3} \pi r^3 \rho \bar{x} + \frac{1}{3} \pi r^3 \rho \times \frac{11r}{4} = 3\pi r^3 \rho \times \frac{3r}{2}$$

correct combination

M1

at least 2 weight $\times$ dist correct

A1

$$\frac{8\bar{x}}{3} = \frac{9r}{2} - \frac{11r}{12}$$

$$\Rightarrow 32\bar{x} = 54r - 11r$$

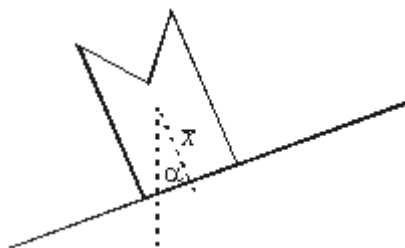
$$\Rightarrow \bar{x} = \frac{43r}{32}$$

A1

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6

(b)



Body will topple when $\alpha > \tan^{-1} \frac{r}{x}$

M1

$$\alpha > \frac{\tan^{-1} \frac{r}{x}}{\frac{43r}{32}}$$

A1f.t.

$$> \tan^{-1} \frac{32}{43}$$

$$> 36.7^\circ$$

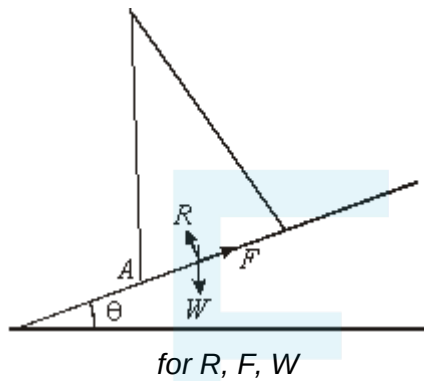
A1f.t.

3

[9]

Q7.

(a)



for R, F, W

for concurrence shown
B0 if weight shown starting from centre of base

B1

B1

2

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(b)



$$\text{C.G. is } \frac{1}{4} \times 6a \text{ from base} = \frac{3a}{2}$$

B1

$$\text{cone topples if } \theta > \frac{\tan^{-1} \alpha}{\frac{3a}{2}}$$

for understanding line of action of W passes through A . May be shown on a diagram

M1

$$> \tan^{-1} \frac{2}{3} \quad (= 33.7^\circ)$$

A1

3

- (c) either $R = W \cos \theta$
or solve

M1

$$F = W \sin \theta$$

both correct

A1

$$F \leq \mu R$$

M1

$$\therefore W \sin \theta \leq \mu W \cos \theta$$

A1

$$\Rightarrow \tan \theta \leq 0.6$$

$$\theta \leq 31.0^\circ$$

A1

$$(\text{or } \tan^{-1} 0.6 < \tan^{-1} \frac{2}{3})$$

$\therefore$ The cone will slide before it topples
ft on c's θ if calculated

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E1F

6

Alternative:

$$F = W \sin \theta$$

M1 A1

$$R = W \cos \theta$$

A1

$$\text{slips when } \frac{F}{R} = \mu$$

M1

$$\text{i.e. } \tan \theta = 0.6$$

A1

$$< \frac{2}{3}$$

slides first

E1

(6)

[11]

Q8.

$$(a) \quad V = \pi \int_0^2 (2 - x) dx$$

Use of $\int x^2 dx$

M1

Correct expression for the volume

A1

$$= \pi \left[2x - \frac{x^2}{2} \right]$$

$$= \pi (4 - 2) = 2\pi$$

ag Correct volume from correct working

A1

3

$$(b) \quad 2\pi \bar{x} = \pi \int_0^2 (x - x^2) dx$$

Use of $\int x^2 dx$

M1

Correct expression containing $\bar{x}$

A1

$$= \pi \left[x^2 - \frac{x^3}{3} \right]_0^2$$

Evaluating integral

M1

$$= \pi \left(4 - \frac{8}{3} \right)$$

$$\bar{x} = \frac{2}{3}$$

Correct final answer

A1

4

(c) $\tan \alpha = \frac{\frac{2}{3}}{\sqrt{2}}$

Use of $\sqrt{2}$

B1

Use of $\tan$ to find angle

M1

Correct expression for $\tan$

A1

$$\alpha = 25.2^\circ$$

Correct angle

Follow through from part (b)

A1ft

4

[11]

Q9.

(a)
$$\bar{x} = \frac{\int_0^5 x(0.4x)^2 dx}{\int_0^5 (0.4x)^2 dx}$$

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Use of appropriate expression for $\bar{x}$

M1

Correct expression excluding limits

A1

Correct limits of integration

A1

$$= \frac{\int_0^5 x^3 dx}{\int_0^5 x^2 dx}$$

$$= \frac{625/4}{125/3} = \frac{15}{4}$$

Evaluation of integrals

m1

ag Correct result from correct working
using both limits

A1

5

(b) Radius = 2

Radius of face = 2

B1

$$\tan \alpha = \frac{2}{\frac{5}{4}} = \frac{8}{5}$$

Use of tan

M1

Use of $5 - \bar{x}$

M1

Correct expression for $\tan \alpha$

A1

$$\alpha = \tan^{-1}\left(\frac{8}{5}\right) = 58.0^\circ$$

Correct angle; allow 58°

A1

5

[10]

EXAM PAPERS PRACTICE

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Q10.

(a) If $\bar{x}$ is the distance of the centre of mass from the base
Moments about plane of the base

M1

$$\pi r^2 \cdot 2r \cdot r \cdot \rho + \frac{1}{3} \pi r^2 H (2r + \frac{H}{4}) \cdot 3\rho$$

M1A1

$$= (\pi r^2 \cdot 2r \rho + \frac{1}{3} \pi r^2 H \cdot 3\rho) \bar{x}$$

A1

$$2r^2 + 2Hr + \frac{H^2}{4} = (2r + H) \bar{x}$$

Dep both M1

m1

$$\bar{x} = \frac{8r^2 + 8Hr + H^2}{4(2r + H)}$$

A1

6

(b) If on the point of toppling

$$\tan \theta = \frac{1}{5} = \frac{r}{\bar{x}}$$

$$\therefore 5r = \frac{8r^2 + 8Hr + H^2}{4(2r + H)}$$

for $\frac{r}{5}$ or $5r =$ result in (a)

M1

A1

$$40r^2 + 20rH = 8r^2 + 8rH + H^2$$

M1

$$32r^2 + 12rH - H^2 = 0$$

$$H^2 - 12rH - 32r^2 = 0$$

A1

$$H = \frac{12 \pm \sqrt{144 + 128}}{2} r$$

$$= (6 \pm 2\sqrt{17})r$$

$$= (6 + 2\sqrt{17})r$$

A1

5

[11]

Examiner reports

Q2.

There was an improved response on this topic than in the past as candidates have clearly learnt the required formula. Integration was generally successful. Some candidates chose to ignore the given volume and recalculated it. The use of π was generally consistent although occasionally got left in the final answer. Many candidates scored full marks on this part. Part (b)(i) was well done and where candidates did not split P into components they did correctly use a compound trig expression which was expanded correctly using an identity from core 4. Part (b)(ii) and (iii) proved more challenging but a significant number of candidates scored full marks here too.

Q3.

The majority of candidates were aware of the required formulae for part (a). Candidates should always state the full formulae first before cancelling the π . In part (b)(i) candidates were aware of the three forces but failed to have the forces all passing through one point. In part (b)(ii) most candidates scored at least two marks; quite often the r/d ratio was incorrect. Part (c) was well understood, but candidates needed to make an explicit comparison for full marks to be awarded.

Q4.

There was a very pleasing response to this question. All marks were lost in part (a) if a two-dimensional formula was used. The best solutions in part (b) used a tabular approach before setting up the relevant equation. A common error was to have the centre of mass of the cone at a distance of $3.5r$ from the base, instead of $2.5r$. The last part was well understood with the correct principle applied. The best responses included a clear labelled diagram showing the principle concerned.

Q5.

Topic – Centres of mass by integration. There was a very good response to this question with candidates showing sound knowledge of formulae and good integration skills. Part (a) was always correctly answered. A common error in part (b) was to lose a π symbol resulting in the wrong answer. Candidates showed good knowledge of suspension problems and even if errors were made in part (b) full recovery marks were available.

Q6.

This was a situation where the final body was obtained by subtracting the cone from the cylinder, and this caused candidates difficulties in drawing up an appropriate “moments equation”. With the answer to the first part having been given, the solution to the final “toppling” problem was standard of its type. Candidates should be aware that they are not penalised here if they have not satisfactorily answered the first part of such a question, but go on to use the information given to pick up later marks.

Q7.

Many candidates failed to gain the mark awarded for showing clearly the concurrence property of three forces in equilibrium. A number showed the weight of the cone acting as if from the centre of the base, rather than the centre of gravity. Parts (b) and (c) required clear thinking and logical argument. This proved difficult for the weaker candidates.

Q8.

There were quite a number of good solutions to this question, but there were also several candidates who used the wrong method. For example some used the method for finding the centre of mass of a uniform lamina. A few candidates had problems squaring the equation of the curve.

In part (c), some candidates did not realise that the curve intersected the y -axis at $\sqrt{2}$.

Q9.

In part (a), the candidates tended to score either zero marks or full marks. The candidates who knew which approach to take were generally able to complete their

$$\bar{x} = \frac{\int xy \, dx}{\int y \, dx}.$$

solutions. A number of candidates used $\int xy \, dx$. Some candidates made errors in their integration and some others made errors when squaring $0.4x$, typically obtaining $0.4x^2$. Part (b) was much more demanding with very few correct answers, but a lot of candidates picked up one or two marks on this part of the question. The two main issues were that candidates did not find the radius of the cone and also that they did not find the distance of the centre of mass from the base of the cone.

Q10.

Better candidates made good progress with this question. In part (a) problems encountered by other candidates included finding the volume of a cone, in which the

third in the formula $\frac{1}{3} \pi r^2 h$ was omitted or was incorrect; the mass of the cone, in which the density 3ρ was not used; and finding the position of the centre of mass of the cone. Part (b) was answered well by many candidates.